CARS DATASET

Exploratory Data Analysis (EDA) Project Report

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Batch:October batch (B)

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Dataset Source:Kaggle

1. Objective

Theobjectiveof this project is to perform exploratory data analysis on a car dataset using Python. EDA helps

understand the structure and quality of the data, identify unusual values, summarize important variables, and explore relationships that may be useful for later analysis or prediction.

Main questions for analysis

- How are car prices distributed?
- Are there unusual values in price and kilometres driven?
- How does a car's model year relate to its price?
- What data-quality issues should be checked before

drawing conclusions?

3. Dataset Overview

First 5 Rows and column of the dataset using df.head() and df.tail()

	car_id	brand	model	year	segment	body_type	fuel_type	transmission	engine_cc	power_hp	fuel_economy_mpg	seats	km_driven	num_owners	condition
0	1	Hyundai	i20	2012	Hatchback	Hatchback	Diesel	Manual	1320	78	42.0	5	120639	1	Good
1	2	Volkswagen	Polo	2021	Hatchback	Hatchback	Petrol	Automatic	1000	116	35.9	5	39084	1	Good
2	3	Volkswagen	Golf	2020	Hatchback	Hatchback	Petrol	Automatic	1400	136	32.7	5	39194	1	Good
3	4	Tata	Nexon	2019	SUV	SUV	Diesel	Manual	1320	115	30.5	5	69792	2	Good
4	5	Mahindra	Scorpio	2024	SUV	SUV	Diesel	Manual	2420	148	23.0	7	9464	1	Good

Last 5 rows

	car_id	brand	model	year	segment	body_type	fuel_type	transmission	engine_cc	power_hp	fuel_economy_mpg	seats	km_driven	num_owners	condition	a
175	176	Chevrolet	Malibu	2021	Midsize	Sedan	Petrol	Automatic	1500	154	29.9	5	39829	2	Excellent	
176	177	Chevrolet	Silverado	2013	Truck	Pickup	Diesel	Automatic	5830	407	21.9	5	94700	1	Good	
177	178	Nissan	Rogue	2023	SUV	SUV	Petrol	Manual	2500	186	30.7	5	18977	1	Good	
178	179	Hyundai	Tucson	2019	SUV	SUV	CNG	Automatic	2500	181	29.5	5	98586	1	Fair	
179	180	Maruti Suzuki	Ertiga	2023	MPV	MPV	CNG	Automatic	1500	97	25.7	7	29724	1	Fair	

4. Data Quality and Cleaning

Thenotebookchecks duplicate records withdf.duplicated().sum(). This gives the number of rows that are exact duplicates across all columns. The result should be recorded after the notebook is run.

```
df.duplicated().sum()
```

```
np.int64(0)
```

This shows dataset contains 0 duplicate values

Outlier detection and treatment

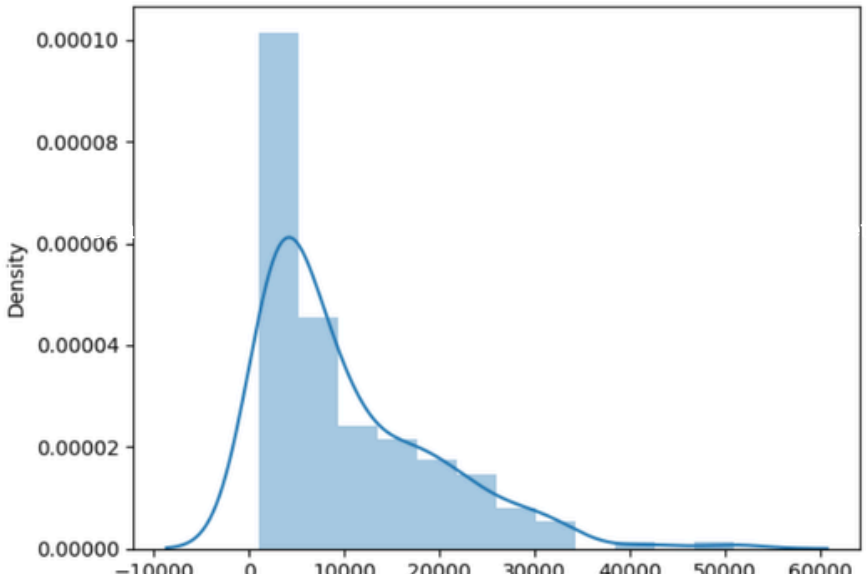
Boxplots are created for price_usd and km_driven to inspect possible outliers. The notebook then applies the Interquartile Range (IQR) method to both columns:

1. Calculate Q1 (25th percentile) and Q3 (75th percentile).
2. Calculate IQR = Q3 – Q1.
3. Calculate lower limit = Q1 – 1.5 × IQR and upper limit = Q3 + 1.5 × IQR.
4. Replace values outside these limits with the nearest limit (capping / winsorisation).

Duplicate rows	Output of df.duplicated().sum()
Missing values	Use df.isnull().sum() and record columns with missing data
Data types	Check df.info(); convert types only when justified
Outliers	Inspect boxplots for price_usd and km_driven
Treatment	IQR-based capping applied to price_usd and km_driven

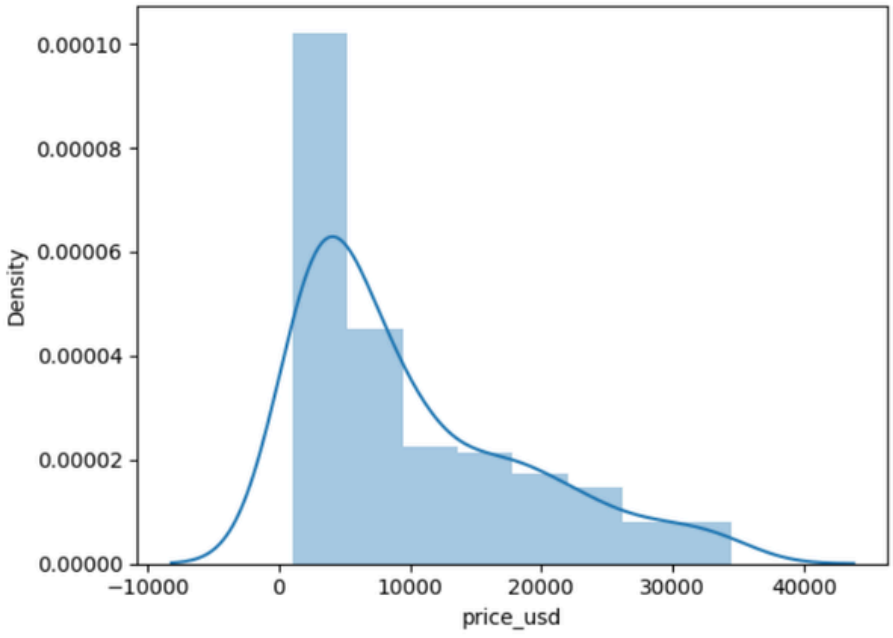
Before outlier treatment

```
sns.distplot(df['price_usd'])
plt.show()
```



After outlier treatment

```
sns.distplot(df['price_usd'])
plt.show()
```



7. Key Findings

1. Prices are right-skewed. Most cars sell for under about 10,000 USD, while a few premium cars reach 42,000 to 51,000 USD (Chart 1).
2. Age is the strongest driver of price. Price has a correlation of +0.75 with year and -0.75 with car age, so newer cars are worth clearly more (Chart 5).
3. Higher mileage goes with lower prices. km_driven has a correlation of -0.64 with price, and high-km cars cluster at low prices (Charts 3 and 5). Mileage and age are closely linked (0.86), so their effects cannot be fully separated here.
4. Automatic cars carry a price premium. Their median price (about 8,100 USD) is roughly 1.7 times that of manual cars (about 4,700 USD) (Chart 4). This is an association; automatic cars may also be newer or more powerful.
5. Power matters more than engine size for price. power_hp (0.60) correlates more strongly with price than engine_cc (0.45) (Chart 5).
6. Fuel type shows a modest price difference. Petrol, Diesel and Hybrid cars average about 10,400 to 10,600 USD, with CNG and Electric lower (Chart 6). Diesel cars do not show a higher average than petrol cars in this data.
7. The brand mix is balanced. Volkswagen, Maruti, Chevrolet and Ford are the most frequent brands, with no single brand dominating (Chart 2).

Limitations of the dataset

- Only 180 records, so results for individual brands and models are based on small counts.
- Electric, Hybrid and CNG cars appear only a few times, so fuel-type averages for them are unreliable.
- No information on service history, tyre or battery condition, or market demand, all of which strongly affect real-world prices.
- Prices are given in USD while the listing cities are Indian cities, so the currency basis should be verified before drawing market conclusions.
- The analysis shows associations, not causes (for example, transmission and price).

8. Conclusion

Thisproject demonstrates a basic EDA workflow: load the data, inspect its structure, check duplicate rows,

visualise numerical variables, detect outliers using boxplots and the IQR rule, cap extreme values, and explore the relationship between model year and car price. The final conclusion should be updated after running the notebook with the actual dataset and inserting the resulting statistics and charts.

Future improvements

Add a missing-value summary, include descriptive statistics in a table, label all chart axes, add at least six exported plots, and compare findings before and after cleaning. If the dataset contains useful fields such as brand, fuel type, or transmission, analyse their distributions and relationship with price as well.